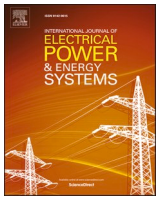
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A novel multimodal framework for ultra-short-term photovoltaic power forecasting via fusion of sky images and time-series data☆

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ABSTRACT

Accurate forecasting of photovoltaic (PV) power generation is of great importance for reducing the uncertainty of distributed PV systems in power grid scheduling and enhancing operational safety and stability. Under variable weather conditions, PV output experiences significant fluctuations caused by changes in cloud patterns in the sky and the combined effects of multiple meteorological factors, thereby increasing the complexity of forecasting. To address these challenges, this study proposes a multimodal PV power forecasting framework, denoted as SFT, that integrates sky images and meteorological time-series. The framework consists of an improved Swin Transformer-based image encoder, a Temporal Fusion Transformer (TFT) branch for time-series modeling, and a gated-attention fusion module. The time-series branch encodes static, meteorological and PV power data, while the image branch converts sky images into numerical feature sequences using physics-based methods and introduces a one-dimensional Swin Transformer to learn deep representations. The fusion module adopts gated-attention to adaptively reweight and integrate the two modalities, ultimately outputting the PV power forecasts. Experimental results show that the proposed method achieves a mean absolute error (MAE) of 0.580 kW in the 10-minute forecasting task, significantly outperforming baseline models and demonstrating superior accuracy and robustness.

1. Introduction

Against the backdrop of intensifying energy crises and climate warming, renewable energy has emerged as the core driver of the energy transition [1]. Photovoltaics (PV) serve as a vital pillar due to their clean, sustainable, and scalable nature [2]. However, their output is intermittent and fluctuates due to weather, cloud cover, irradiance, and equipment [3], posing challenges for grid stability, dispatch, and management. Especially in distributed small-scale arrays, accurate ultra-short-term power forecasting is crucial for efficient microgrid operation [4]. Data-driven approaches have gained attention for significantly improving accuracy by integrating multimodal data [5].

Traditional PV power forecasting research has primarily relied on time-series modeling, gradually incorporating external information such as remote sensing and weather forecasts to forecast future variations [6]. Shadab et al. employed a seasonal autoregressive integrated moving average (SARIMA) model to simulate and forecast NASA POWER remote sensing monthly irradiance sequences, generating spatial distributions of future monthly average irradiance for the study area. This enabled the identification of potential high-irradiance development zones [7]. Eseye et al. proposed a hybrid model integrating support vector machine (SVM) and other algorithms, combining power measured by supervisory control and data acquisition (SCADA) with numerical weather prediction (NWP) meteorological data for intraday PV power forecasting. Experiments validated that this hybrid framework enhances forecasting accuracy [8]. Abumohsen et al. introduced an ensemble model based on random forest (RF) for solar power generation time series forecasting, reporting high accuracy for this task [9]. However, single-modal modeling has limited information sources, struggles to fully account for scenarios like shading and rapid cloud cover changes, and suffers from constrained generalization capabilities. Multi-source data does not inherently guarantee improvement; inadequate feature extraction and fusion can introduce redundant noise and increase complexity, potentially degrading performance [10]. Relying solely on numerical time-

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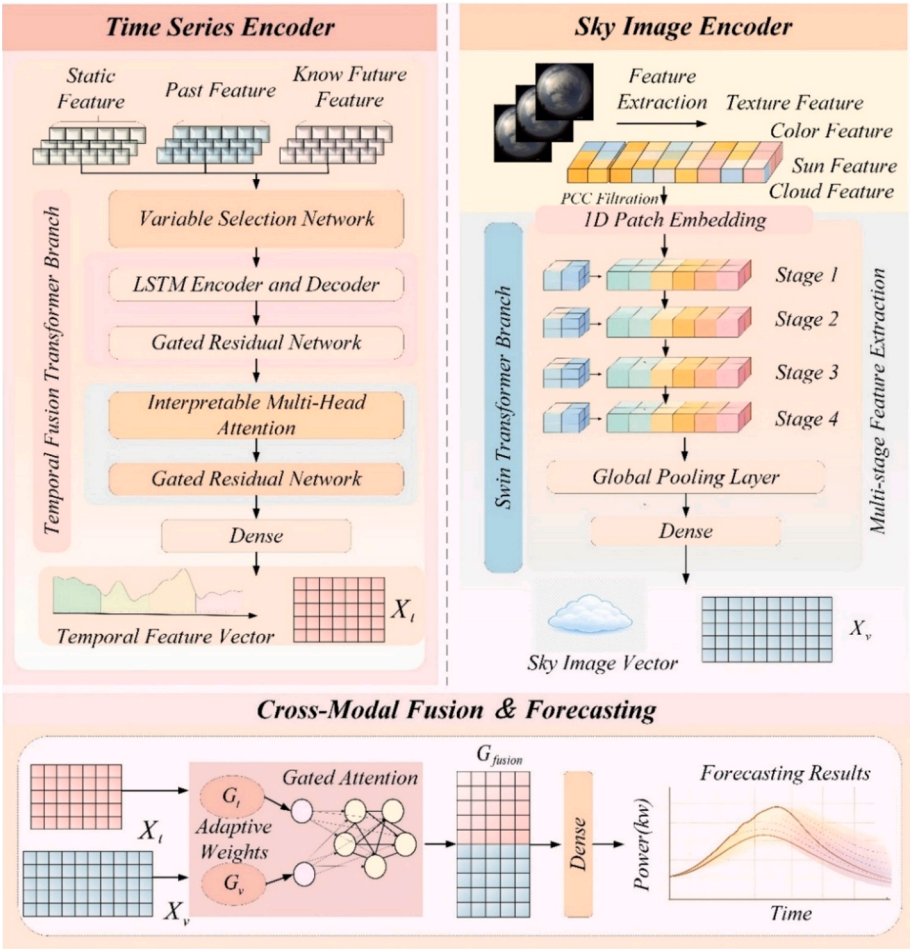


Fig. 1. Overall architecture of the multimodal fusion framework integrating Swin Transformer and Temporal Fusion Transformer (TFT) for PV power forecasting.

series data may neglect critical information contained in sky images, making it difficult to capture short-term fluctuations [11].

Deep learning offers new possibilities for PV forecasting [12]. The convolutional neural network (CNN) can extract spatial features from sky images to characterize cloud structure dynamics [13]. Recurrent neural network (RNN) models and their variants—long short-term memory (LSTM) and gated recurrent unit (GRU)—are adept at modeling long-term dependencies in meteorological time-series [14,15]. However, using CNN or RNN alone has limitations in multimodal fusion, struggling to simultaneously integrate images and meteorological time-series [16]. Existing methods still exhibit unstable performance under rapidly changing cloud conditions [17].

Transformer-based models have been widely applied in tasks such as natural language processing, computer vision, and time-series forecasting [18]. The self-attention mechanism facilitates processing multi-source inputs within a single- model, providing a unified modeling approach for multimodal forecasting [19]. In forecasting tasks, the Temporal Fusion Transformer (TFT) can simultaneously incorporate multiple variable sources while reducing the impact of irrelevant information through gating and variable selection [20]. Nazir et al. applied TFT to forecasting energy consumption demand for smart grid users, achieving results superior to baseline models like LSTM and temporal convolutional network (TCN) [21].

In image learning, the Swin Transformer employs shifted window self-attention to capture local details and cross-region information, finding broad application across diverse visual tasks [22]. For PV power forecasting, Xu et al. proposed a transferable forecasting framework based on Swin Transformer using panoramic sky camera images, enabling ultra-short-term PV power forecasting for new stations with limited generation data [23]. Beyond 2D visual tasks, shifted window self-attention has also been adapted for 1D sequence modeling. For instance, Li et al. introduced the 1D Swin Transformer for genome sequence-related prediction tasks [24]. These works demonstrate that the Swin Transformer's window mechanism is applicable not only to image feature learning but also to modeling one-dimensional signals in long sequences.

For ultra-short-term forecasting of PV power and solar irradiance, existing research has combined sky image and historical numerical sequence modeling to enhance forecasting accuracy [25]. A common approach involves setting up separate feature extraction branches for images and sequences, then introducing mechanisms like attention in a fusion layer for joint learning and forecasting [26]. This “cross-modal fusion framework” typically entails using a visual model for the image side and a time-series model for the sequence side, culminating in joint forecasting within a fusion module. Many existing studies primarily use single-source historical time-series as input, with relatively limited incorporation of historical sequences and NWP data [27]. Moreover, most approaches encode two-dimensional images directly; in contrast, the pipeline that first converts sky images into continuous one-dimensional numerical feature sequences and then performs forecasting using 1D-adapted image architectures remains insufficiently investigated.

To address the aforementioned issues, this study proposes a multimodal fusion method for ultra-short-term power forecasting, denoted as SFT. This approach employs a Transformer-based model (TFT) to jointly model static, meteorological and PV power data, thereby extracting effective information related to power variations. Sky images undergo physical processing to extract numerical features, forming continuous

feature sequences. A 1D-adapted Swin Transformer is then used to model temporal variation patterns in these sequences, capturing how cloud evolution drives short-term power fluctuations. During the fusion stage, gated-attention is introduced to adaptively weight the two feature types. Finally, a fully connected (dense) layer provides the forecast output, enhancing forecasting accuracy and stability under complex weather conditions. Unlike existing studies that directly encode raw sky images or apply simple multimodal concatenation, this work converts sky-image information into physically meaningful temporal feature sequences and integrates it with TFT-based time-series representations through gated-attention fusion. The main contributions of this study are as follows:

(1) A multimodal fusion model for ultra-short-term power forecasting of small-scale PV arrays is proposed, which differs from conventional multimodal methods by representing sky-image information as temporal feature sequences rather than directly encoding raw 2D images.
(2) Static plant information, historical power, and NWP-based known future covariates are jointly modeled within a unified time-series forecasting framework using TFT, while the influence of irrelevant features is adaptively suppressed through variable selection and gated fusion mechanisms.
(3) The hierarchical Swin Transformer is adapted into a one-dimensional temporal architecture to model sequences of sky-image-derived features, enabling window-based and shifted-window self-attention to capture temporal dependencies associated with cloud dynamics.
(4) A fusion method based on gated-attention is proposed to achieve adaptive weighting of the two feature types. Extensive experiments validate the method's effectiveness under complex weather conditions, particularly during rapid cloud changes.

2. Methodology

The overall framework of the proposed SFT is shown in Fig. 1. The proposed model adopts a multimodal architecture that integrates sky images and time-series data. The time-series branch employs a Temporal Fusion Transformer (TFT) to capture long-term dependencies in meteorological data, while the image branch encodes sky images into numerical feature sequences and then applies a 2D-to-1D adapted Swin Transformer to extract deep features. Feature representations generated by these two branches are subsequently fused through a gated-attention mechanism that dynamically assigns modal weights. Finally, a dense layer produces the forecast output. This architecture effectively leverages complementary information from different modalities, significantly improving the accuracy of PV power forecasting, particularly under complex weather conditions.

2.1. PV power denoising and reconstruction using PSO-VMD

To improve the quality of PV power time-series data, this study incorporates a variational mode decomposition (VMD) method optimized via particle swarm optimization (PSO). The approach decomposes and reconstructs the original signal, aiming to reduce noise interference and enhance the stability of the resulting series.

In the VMD framework, the input signal $x(t)$ is decomposed into K mode components $u_k(t)$, each of which corresponds to a specific center frequency ω_k.

The core objective of this decomposition process is to minimize the sum of the estimated bandwidths of all modes.

This can be mathematically formulated as the following variational optimization problem [28]:

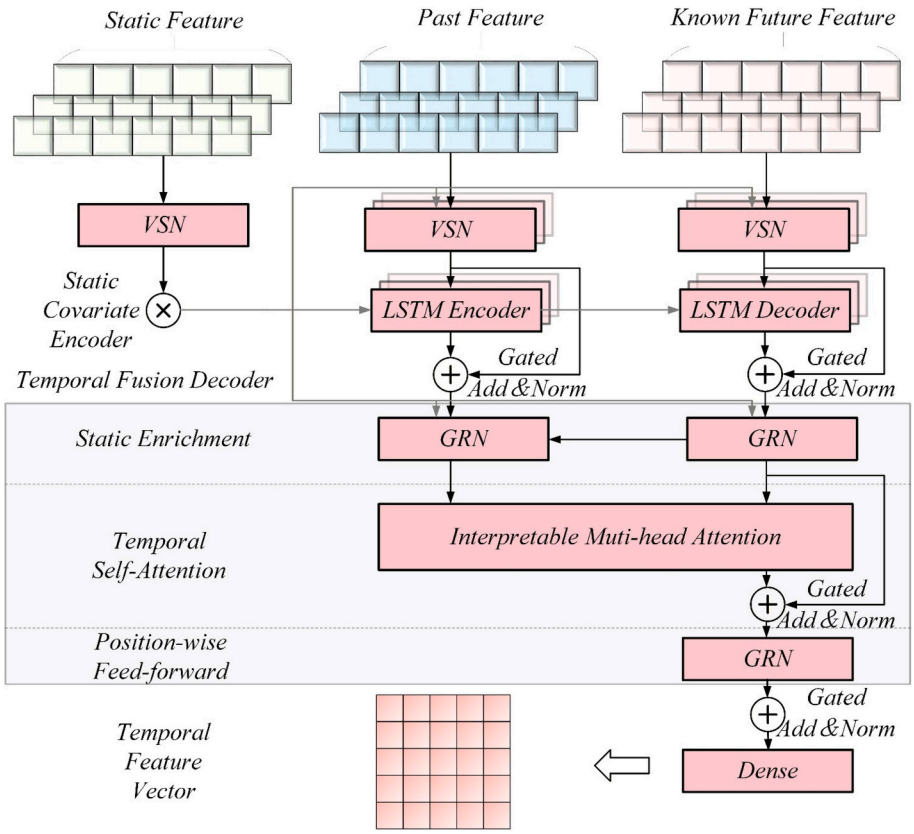


Fig. 2. Architecture of the TFT with variable selection and static feature integration for PV power forecasting.

$$\min_{\{u_k\},\{\omega_k\}}\left\{\sum_{k=1}^{K}\left\|\partial_t\left[\left(\delta(t)+\frac{j}{\pi t}\right)*u_k(t)\right]e^{-j\omega_k t}\right\|_2^2 + \left\|x(t)-\sum_{k=1}^{K}u_k(t)\right\|_2^2\right\} \tag{1}$$

where α denotes the bandwidth parameter, ∂_t denotes the he partial derivative with respect to time t, j denotes the imaginary unit, $*$ denotes the convolution operation and $\delta(t)$ denotes the Dirac distribution.

By applying the alternating direction method of multipliers (ADMM) in the frequency domain to iteratively solve the variational optimization problem in equation (1), one can obtain the modes u_k, center frequencies ω_k , and the update formula for the Lagrange multiplier λ:

$$\begin{aligned}\widehat{u}_k^{n+1}(\omega) &= \frac{\widehat{f}(\omega)-\sum i\neq k\widehat{u}_i^n(\omega)+\frac{\widehat{\lambda}^n(\omega)}{2}}{1+2\alpha(\omega-\omega_k^n)^2}\\ \omega_k^{n+1} &= \frac{\int_0^\infty \omega\left|\widehat{u}_k^{n+1}(\omega)\right|^2 d\omega}{\int_0^\infty \left|\widehat{u}_k^{n+1}(\omega)\right|^2 d\omega}\\ \widehat{\lambda}^{n+1}(\omega) &= \widehat{\lambda}^n(\omega)+\tau\left(\widehat{f}(\omega)-\sum_{k=1}^{K}\widehat{u}_k^{n+1}(\omega)\right)\end{aligned} \tag{2}$$

where $\widehat{u}(\omega)$ denotes the intrinsic mode function after performing the Fourier transform, $\widehat{f}(\omega)$ denotes the Fourier transform of the original input signal $x(t)$, $\widehat{\lambda}(\omega)$ denotes the Fourier transform of the Lagrange multiplier, ω denotes the central frequency of each mode, τ the step size used to control the update speed of the Lagrange multiplier, and n denotes the number of iterations. PSO is employed to search for α and τ to ensure the optimality of the decomposed mode u_k and the central frequency ω_k.

This study employs PSO to optimize the parameters and iteratively solves for the optimal modal number. This approach thereby attains superior decomposition and denoising performance. As a heuristic optimization algorithm, PSO identifies the optimal solution by mimicking the foraging behavior of bird flocks:

$$v_{i,m}^{t+1} = w\cdot v_{i,m}^t + c_1\cdot r_1\cdot(pbest_{i,m}-x_{i,m}^t)+c_2\cdot r_2\cdot(gbest_m-x_{i,m}^t) \tag{3}$$

where $v_{i,m}$ denotes the velocity of the $i-$th particle in the $m-$th dimension at iteration t, $x_{i,m}^t$ denotes the position of the $i-$th particle in the $m-$th dimension at iteration t, w denotes the inertia weight, c_1 and c_2 denote the cognitive and social learning factors, r_1 and r_2 denote random numbers in the range [0, 1], $pbest_{i,m}$ denotes the best position found so far by the $i-$th particle, and $gbest_m$ denotes the best position found so far by the entire swarm in the $m-$th dimension.

The particle position update formula is:

$$x_{i,m}^{t+1} = x_{i,m}^t + v_{i,m}^{t+1} \tag{4}$$

These two formulas drive the particle swarm to search within the solution space until convergence to the global optimum. They effectively reduce the non-stationarity of the original data, serving as input for subsequent time-series modules.

2.2. TFT for Time-Series feature encoding

This study employs the TFT framework proposed by Lim et al.[29], with the overall TFT framework is shown in Fig. 2. This framework is specifically designed for multidimensional time-series. The model's input encompasses historical dynamic features, known future dynamic features, and static features. For dynamic features, temporal information is extracted using a variable selection network, LSTM, and gated residual networks. Static features are encoded into context vectors to regulate variable selection and initialize the LSTM. After integrating these components, an interpretable multi-head attention mechanism captures key dependencies. The processed data then performs quantile regression to generate forecasts.

2.2.1. Gating mechanism

In the TFT structure, the function of the gating mechanism is to selectively apply nonlinear transformations to specific input features, while providing a direct passage for some features to bypass nonlinear processing. This helps avoid unnecessary computation on features that contribute little or contain redundant information. The gated residual network (GRN) is its basic component, which takes the main input $\boldsymbol{a}$ and an optional context vector $\boldsymbol{c}$ as inputs where $\boldsymbol{a}$ denotes the current feature representation processed by the GRN, and $\boldsymbol{c}$ denotes the context vector generated by the static covariate encoder for conditioning and modulation.

$$\mathrm{GRN}_\omega(\boldsymbol{a},\boldsymbol{c}) = \mathrm{LN}(\boldsymbol{a}+\mathrm{GLU}_\omega(\eta_1)) \tag{5}$$

$$\eta_1 = W_{1,\omega}\eta_2 + b_{1,\omega} \tag{6}$$

$$\eta_2 = \mathrm{ELU}\left(W_{2,\omega}\boldsymbol{a}+W_{3,\omega}\boldsymbol{c}+b_{2,\omega}\right) \tag{7}$$

$$\mathrm{GLU}_\omega(\gamma) = \sigma\left(W_{4,\omega}\gamma+b_{4,\omega}\right)\odot\left(W_{5,\omega}\gamma+b_{5,\omega}\right) \tag{8}$$

where η_1 and η_2 denote the hidden layer variables,$W_{i,\omega}$ and $b_{i,\omega}$ denote the weights and biases of the $\omega-$th submodule, LN denotes layer normalization, *ELU* denotes the exponential linear unit activation function, $\sigma(\cdot)$ denotes the Sigmoid activation function, γ denotes the input to the GLU, and $\odot$ denotes the element-wise product. $GLU(\cdot)$ denotes the gated linear unit, which combines the gating signal and feature values from the sigmoid function to flexibly control the information flow. By adjusting η_1, the contribution ratio of each module to the final output can be tuned, thereby controlling the influence of nonlinear transformations on the final result.

2.2.2. Variable selection network

The variable selection network *VSN* aims to identify key features while suppressing noise. After embedding and transforming both static and dynamic variables, let $\xi_i^{(j)}$ denote the embedded vector of the $j-$th input variable at time t. This vector is then mapped nonlinearly through the GRN to obtain the feature vector $\widetilde{\xi}_t^{(j)}$.

$$\widetilde{\xi}_t^{(j)} = GRN_{\tilde{\xi}^{(j)}}\left(\xi_t^{(j)}\right) \tag{9}$$

The embeddings of all historical input variables are concatenated sequentially to form the flattened input vector:

$$\boldsymbol{\Xi}_t = \left[\xi_t^{(1)},\ldots,\xi_t^{(m_x)}\right]^\top \tag{10}$$

where m_x denotes the number of input variables. Combining the external static context vector $\boldsymbol{C}_s$, the selection weights for each variable are computed via a second GRN and normalized using the softmax function:

$$v_{\chi_t} = \mathrm{Softmax}\left(GRN_{v_\chi}(\boldsymbol{\Xi}_t,\boldsymbol{C}_s)\right) \tag{11}$$

where v_{χ_t} denotes a vector of variable selection weights. Finally, the fused features for the current step are generated through weighted summation:

$$\widetilde{\xi}_t = \sum_{j=1}^{m_y} v_{\chi_t}^{(j)}\widetilde{\xi}_t^{(j)} \tag{12}$$

where $v_{\chi_t}^{(j)}$ denote the $j-$th element of vector v_{χ_t}.

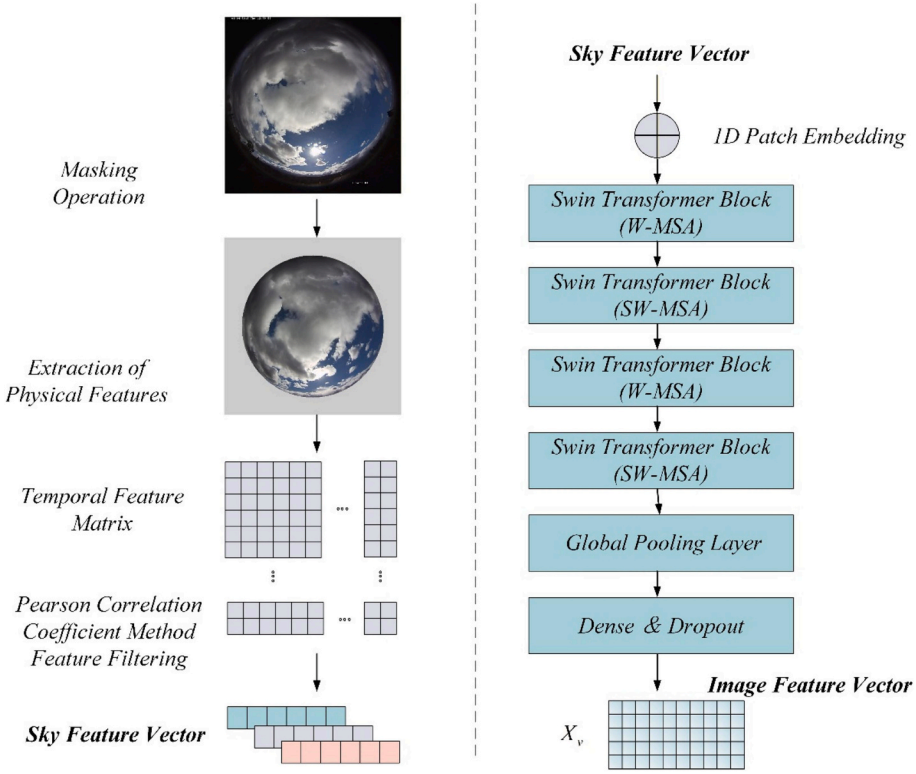


Fig. 3. Framework of the sky image branch for PV power forecasting.

2.2.3. Static covariate encoder

The static covariate encoder transforms non-time-dependent static inputs S_d into static embeddings and generates context vectors to modulate other model components. First, static inputs are embedded as:

$$\boldsymbol{S}' = \mathrm{ReLU}(W_2 \cdot \mathrm{ReLU}(W_1 S_d + b_1) + b_2) \tag{13}$$

Subsequently fed into the static GRN, it outputs four types of context vectors:

$$(\boldsymbol{C}_s, \boldsymbol{C}_e, \boldsymbol{C}_c, \boldsymbol{C}_h) = \mathrm{GRN}_{\mathrm{static}}(\boldsymbol{S}') \tag{14}$$

where $\boldsymbol{C}_s$ provides contextual information for the VSN to perform feature selection, $\boldsymbol{C}_e$ represents the temporal features used to enhance the encoding–decoding process, and $\boldsymbol{C}_c$ and $\boldsymbol{C}_h$ are used to initialize the cell state and hidden state of the LSTM encoder, respectively, enabling static information to influence the initialization of the sequence model.

2.2.4. Interpretable multi-head attention

In the TFT model, an interpretable multi-head attention mechanism is introduced to more effectively capture long-term dependencies within sequences. Based on the framework proposed by Vaswani et al. [30], the single-head scaled dot-product attention is defined as:

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^{\mathrm{T}}}{\sqrt{d_k}}\right)V \tag{15}$$

where Q, K, and V denote the query, key, and value matrices, respectively, and d_k denotes the dimension of the key vector.

Traditional multi-head mechanisms assemble results from each head through parallel computation:

$$\mathrm{MultiHead}(Q, K, V) = \left[H_1, \ldots, H_{N_h}\right] W^O \tag{16}$$

$$H_h = \mathrm{Attention}(QW_h^Q, KW_h^K, VW_h^V) \tag{17}$$

where W_h^Q,W_h^K,and W_h^V denote the projection matrices for the $h-$th attention head, W^O denotes the output projection matrix, and N_h denotes the number of attention heads.

TFT improves the output of each head by performing mean aggregation to obtain a unified timing representation:

$$\widetilde{H} = \frac{1}{N_h}\sum_{h=1}^{N_h} \mathrm{Attention}(QW_h^Q, KW_h^K, VW_h^V) \tag{18}$$

$$\mathrm{InterpretableMultiHead}(Q, K, V) = \widetilde{H}W^O \tag{19}$$

where $\widetilde{H}$ denotes the averaged attention representation across all heads. This aggregation method reduces differences between heads, facilitating analysis of attention distributions across time steps and thereby effectively enhancing model interpretability.

2.2.5. Quantile regression output layer

To provide probabilistic forecasts, TFT employs quantile regression in its output layer. Let y_t denote the ground-truth target value at time (t). Given the input information available up to time (t), TFT outputs multi-step quantile forecasts $\hat{y}_{t+h|t}^{(q)}$ for $(h = 1, \ldots, H)$ and $(q \in Q)$, where the conditional subscript "$(t+h|t)$" indicates "forecasting the value at time $(t+h)$ conditioned on observations up to time (t)", H denotes the forecasting horizon and Q denotes the set of target quantiles q. The model is trained to simultaneously forecast multiple quantiles, aiming to minimize the joint quantile loss across all time steps and target quantiles:

$$L(\Omega, \boldsymbol{W}) = \sum_{y_t \in \Omega}\sum_{q \in Q}\sum_{h=1}^{H} \frac{QL\left(y_{t+h}, \hat{y}_{t+h|t}^{(q)}, q\right)}{MH} \tag{20}$$

$$QL(y, \hat{y}, q) = q(y - \hat{y})_+ + (1-q)(\hat{y} - y)_+ \tag{21}$$

where Ω denotes the training dataset containing M samples, $\boldsymbol{W}$ denotes the weights of the TFT, and $(\bullet)_+ = \max(0, .)$. This mechanism models the uncertainty of multiple quantile forecasts; during point forecast, the

median quantile is typically selected. After training, the median quantile ($q = 0.5$) is adopted to obtain the multi-step point forecast vector:$X_t \in \mathbb{R}^H$:

$$\widehat{y}_t = \left[\widehat{y}^{(0.5)}_{t+1|t}, \widehat{y}^{(0.5)}_{t+2|t}, \ldots, \widehat{y}^{(0.5)}_{t+H|t}\right]^\top \tag{22}$$

Through a dense layer, we further encode this vector into a compact temporal feature for fusion, denoted as $X_t \in \mathbb{R}^d$, where d denotes the feature dimension.

2.3. Sky image encoder with 1D swin transformer

2.3.1. Physical-statistical feature extraction from sky images

As shown in Fig. 3, sky images undergo feature extraction through physical methods, converting them into numerical feature sequences encompassing color features, cloud features [31], solar features, and texture features [32,33]. Color features include the mean and standard deviation of RGB and HSV channels; cloud features include coverage, type, morphology, and edge complexity; solar features include position and directional cloud cover; texture features include Sobel gradient, local binary pattern (LBP), and gray-level co-occurrence matrix (GLCM). The preprocessing stage employs masks to exclude interference and performs downsampling to optimize efficiency [34]. To eliminate redundant variables, the Pearson correlation coefficient (PCC) is used for screening, retaining only strongly correlated features with a correlation greater than 0.6 with photovoltaic power as model inputs.

2.3.2. One-dimensional swin transformer network architecture

Following the approach of Kao et al. [35], this study adapts the hierarchical architecture and shifted window self-attention mechanism of the Swin Transformer into a 1D temporal feature model to efficiently capture both short-term and long-term dependencies. By modifying the input layer, windowing mechanism, and positional encoding, the model achieves a transition from two-dimensional image processing to one-dimensional temporal analysis.

Unlike two-dimensional image processing, this model first processes the input sequence $X_i \in \mathbb{R}^{B\times L\times C}$ through a one-dimensional patch embedding module, where B denotes the batch size, C denotes the input dimension, and L denotes the sequence length. This module comprises two steps: sequence segmentation and feature projection. It employs a one-dimensional convolutional layer of size P to perform mapping on the sequence, projecting contiguous, non-overlapping patches into high-dimensional embedded features.

For the $r-$th dimension of the feature value of the $k-$th patch in the output feature y, its calculation formula is defined as:

$$y^r_k = b_r + \sum_{i=1}^{C_{\text{in}}}\sum_{j=1}^{P} w_{r,i,j}\cdot x_{(k\times S+j),i} \tag{23}$$

where b_r denotes the convolution bias, C_{in} denotes the number of input channels,S denotes the convolution stride, $w_{r,i,j}$ denotes the convolution kernel weights and x denotes the input sequence after padding processing. This operation effectively converts a one-dimensional time-series signal into a latent space vector that can be interacted with by the model.

The feature extraction core consists of Swin Transformer blocks that alternately apply window-based multi-head attention (W-MSA) and shifted window-based multi-head attention (SW-MSA). At each stage, consecutive Swin Transformer blocks are stacked alternately: the preceding block employs W-MSA, while the subsequent block uses SW-MSA. The forward propagation of each block sequentially performs layer normalization, attention computation, residual connection, another layer normalization, multilayer perceptron (MLP) computation, and a final residual connection. After passing through all the blocks in a stage (except the final stage), a patch merging layer is applied to perform downsampling, and the resulting feature map is then fed into the next stage.

To adapt to one-dimensional data, the token sequences are divided into non-overlapping windows of size M. Leveraging a cyclic shift mechanism to establish cross-window interactions, this model directly embeds one-dimensional temporal relative-position bias into self-attention computations to enhance temporal awareness, as expressed by the following formula:

$$\text{Attention}(Q,K,V) = \text{softmax}\left(\frac{QK^{\text{T}}}{\sqrt{d_k}} + B_v\right)V \tag{24}$$

where Q, K, and V denote the query, key, and value matrices, d_k denotes the feature dimension, and B_v denotes the one-dimensional relative position offset matrix introduced. For a one-dimensional token sequence with window size M, the relative position index range between two patches is $[-M+1, M-1]$. The value of the bias matrix B_v is obtained from the parameterized matrix $\widehat{B} \in \mathbb{R}^{(2M-1)\times N_h}$ based on this index.

The forward propagation process for two consecutive Swin Transformer blocks is as follows:

$$\begin{aligned}
\widehat{z}^l &= \text{W - MSA}(\text{LN}(z^{l-1})) + z^{l-1}\\
z^l &= \text{MLP}(\text{LN}(\widehat{z}^l)) + \widehat{z}^l\\
\widehat{z}^{l+1} &= \text{SW - MSA}(\text{LN}(z^l)) + z^l\\
z^{l+1} &= \text{MLP}(\text{LN}(\widehat{z}^{l+1})) + \widehat{z}^{l+1}
\end{aligned} \tag{25}$$

where z^l and $\widehat{z}^l$ denote the output features of the $l-$th layer's (S)W-MSA module and *MLP* module, respectively, $W-MSA$ denotes the window-based multi-head attention mechanism, $SW-MSA$ denotes the shifted-window multi-head attention mechanism that achieves cross-window feature interaction through cyclic window shifting, and *MLP* denotes the multilayer perceptron module.

Additionally, the model employs a Patch Merging layer for downsampling, halving the token sequence length while doubling the dimensions. This approach constructs hierarchical feature representations and expands the receptive field. After the Swin Transformer blocks, we apply global pooling to summarize the token sequence into a single vector, and then use a dense layer to obtain the compact visual feature $X_v \in \mathbb{R}^d$ for cross-modal fusion, where d is set to be the same as the temporal feature dimension to facilitate fusion.

2.4. Cross-modal fusion module

Inspired by the Gated-Attention Reader of Dhingra et al. [36], which leverages multiplicative interactions to modulate intermediate representations conditioned on another input, this study introduces a gated-attention fusion module to adaptively modulate and combine the outputs of different branches.

First, the features of both modalities are normalized, and the normalized features are then input into their respective trainable linear mapping layers to compute corresponding modal feature score vectors. Transform the original features into a new space via a trainable linear mapping, where each dimension's value reflects its importance within that modality:

$$G_t = W_tX_t + b_t, G_v = W_vX_v + b_v \tag{26}$$

where W_v and W_t denote the weight matrices, b_t and b_v denote the bias vectors, G_t and G_v denote the score vectors for the temporal modality and the image modality, respectively.

The two score vectors are stacked along the modality dimension, with an adjustable temperature parameter T to use to control the weight distribution:

$$[\alpha_t, \alpha_v] = \text{softmax}\left(\frac{[G_t, G_v]}{T}\right) \tag{27}$$

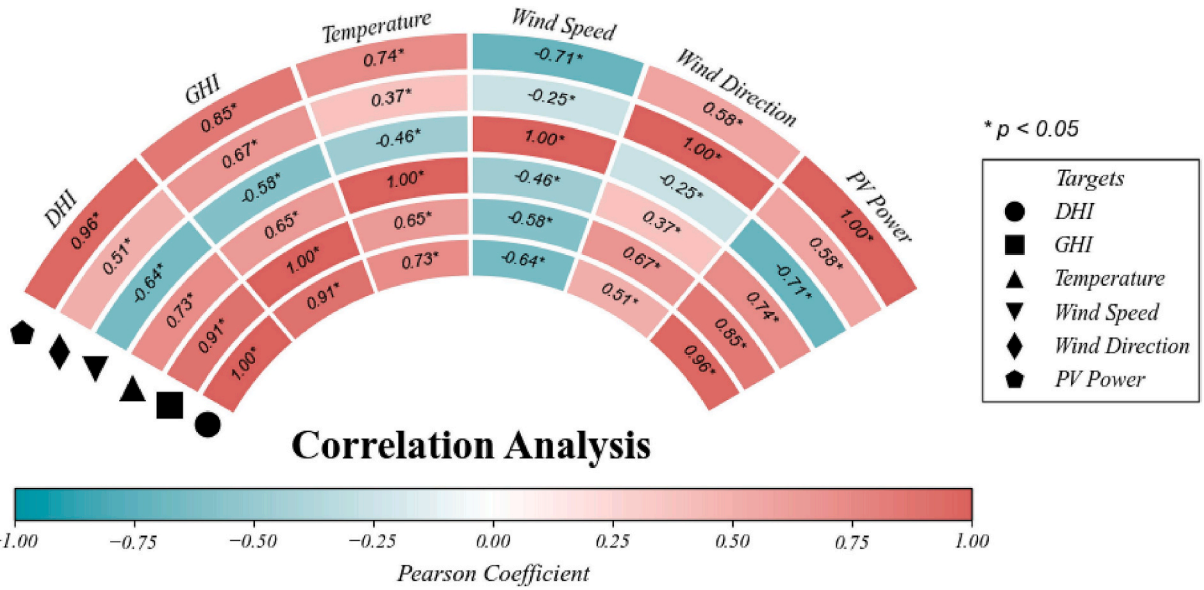


Fig. 4. Heatmap of PCC between meteorological variables and PV power output.

where α_t denotes the temporal weight, and α_v denotes the image weight.

Using the learned weights, the two modality forecasts are adaptively gated via element-wise multiplication:

$$\widetilde{X}_t = \alpha_t \odot X_t \\ \widetilde{X}_v = \alpha_v \odot X_v \tag{28}$$

where $\widetilde{X}_t$ and $\widetilde{X}_v$ respectively denote the temporal modality feature component and the image modality feature component.

After completing gated-attention processing, the gated modality features are concatenated to form a unified fused representation:

$$G_{fusion} = \left[\widetilde{X}_t, \widetilde{X}_v\right] \tag{29}$$

where G_{fusion} denotes the fused multimodal representation. After processing the fused features through a dense layer, the model outputs the forecast value $\widehat{Y} \in \mathbb{R}^H$. This model efficiently integrates multimodal features through a dynamic gating mechanism, significantly enhancing forecasting accuracy—particularly in complex cloud-change scenarios.

3. Experiment

3.1. Dataset

This study utilizes all-sky imagery released by Stanford University in 2018 along with corresponding 30 kW photovoltaic output data. During the preprocessing stage, the imagery underwent downsampling and quality screening, retaining 15,000 valid samples, which were uniformly aligned to a 10-minute temporal resolution. Simultaneously, meteorological records corresponding to the image acquisition locations were extracted from the National Solar Resource Database (NSRDB) established by the National Renewable Energy Laboratory (NREL). The results of the Pearson correlation analysis are shown in Fig. 4. Based on this method, five feature variables significantly correlated with photovoltaic power were ultimately selected: diffuse irradiance (DHI), global irradiance (GHI), air temperature, wind speed, and wind direction. These were used as model inputs for the temporal branching.

3.2. Experimental setup and scheme

The experimental platform for this study utilizes an Intel® Core™ i5-13400F processor, 32 GB of memory, and an NVIDIA GeForce RTX 4060 GPU, running on the Windows 11 operating system. The implementation environment consists of Python 3.9 and the PyTorch framework. Model training uniformly employed the Adam optimizer with an initial learning rate of 0.001 and a batch size of 16, utilizing mixed-precision training and an early stopping mechanism.

Table 1
Key hyperparameter settings of baseline and variant models used in comparative experiments.

Category	Model	Hyperparameters
Time-series Benchmark	GRU	2 layers, 128 hidden units
	LSTM	2 layers, 128 hidden units
	Transformer	3 encoder layers, 2 decoder layers, 8 attention heads.
Image Benchmark	ResNet-18	Residual filters [64,128,256,512], 2 blocks per stage.
	MobileNetV2	Inverted residuals with filters [16, 24(x2), 32 (x3), 64(x4), 96(x3), 160(x3), 320(x1)], expansion factor 6.
	ViT	6 encoder layers, 4 attention heads, model dimension 128.

To comprehensively evaluate model performance and the effectiveness of fusion strategies, this section designs a two-stage comparative experiment. In the first stage, we validate the advantages of multimodal models over unimodal baselines: for the time-series modality, we employ LSTM, GRU [37], Transformer, and TFT; while the image modality employs ResNet18 [38], MobileNetV2 [39], Vision Transformer (ViT) [40], and Swin Transformer to highlight the value of multimodal fusion. For the multidimensional time-series features extracted from sky images using physics-based methods, we reshape them into "pseudo-images" to adapt them to the input requirements of image-modal models [41].

The second stage focuses on evaluating the validity of the gated-attention fusion framework and the choice of backbone networks. By replacing TFT and Swin Transformer with the best-performing single-modal baseline models from the first stage, we validate the framework's adaptability and robustness across different backbones. Simultaneously, ablation experiments are conducted to analyze the performance contributions of each component. Key hyperparameter configurations are listed in Table 1, with all other settings remaining consistent.

Table 2
Optimized VMD parameters obtained via PSO under different mode numbers.

K	α	τ	RMSE (kW)	SNR (dB)
2	99.96	0.012	1.298	21.690
3	213.83	0.016	1.047	23.560
4	281.18	0.015	1.296	21.700
5	489.06	0.026	1.510	20.376
6	483.41	0.004	1.605	19.846
7	494.52	0.003	1.639	19.663

3.3. Evaluation metrics

This study selects mean absolute error (MAE), root mean square error (RMSE), and coefficient of determination (R2) as core evaluation metrics. MAE measures the average absolute deviation between forecasted and actual values, reflecting overall model accuracy while being less sensitive to a few outliers. RMSE incorporates squared processing in error calculation, providing a more sensitive portrayal of extreme deviation scenarios. R2 quantifies the model's ability to explain actual power fluctuations and variations. The mathematical definitions of these metrics are as follows:

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i| \tag{30}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \tag{31}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2} \tag{32}$$

where N denotes the total number of samples, y_i and $\hat{y}_i$ denote the actual value and the forecasted value of the i−th sample, respectively, and $\bar{y}$ denotes the mean value of all samples. Smaller values of MAE and RMSE indicate a smaller deviation from the true values and better model performance, while an R^2 value closer to 1 indicates a higher degree of model fit. In addition, the signal-to-noise ratio (SNR) is used to evaluate the reconstruction quality in the PSO-VMD parameter optimization experiment. It is defined as:

$$SNR = 10\log_{10}\left(\frac{\sum_{t=1}^{N}x(t)^2}{\sum_{t=1}^{N}(x(t) - \hat{x}(t))^2}\right) \tag{33}$$

where $x(t)$ and $\hat{x}(t)$ denote the original signal and the reconstructed signal, respectively. A larger SNR indicates better reconstruction quality.

4. Results

4.1. VMD parameter optimization results and power reconstruction

The optimal parameters for VMD are determined using the PSO algorithm described in Section 2.2.1. Table 2 presents the optimal decomposition results for various modal numbers. The decomposition achieves the best performance when the modal number K was increased to 3, yielding SNR = 23.560 dB and RMSE = 1.047 kW. Further

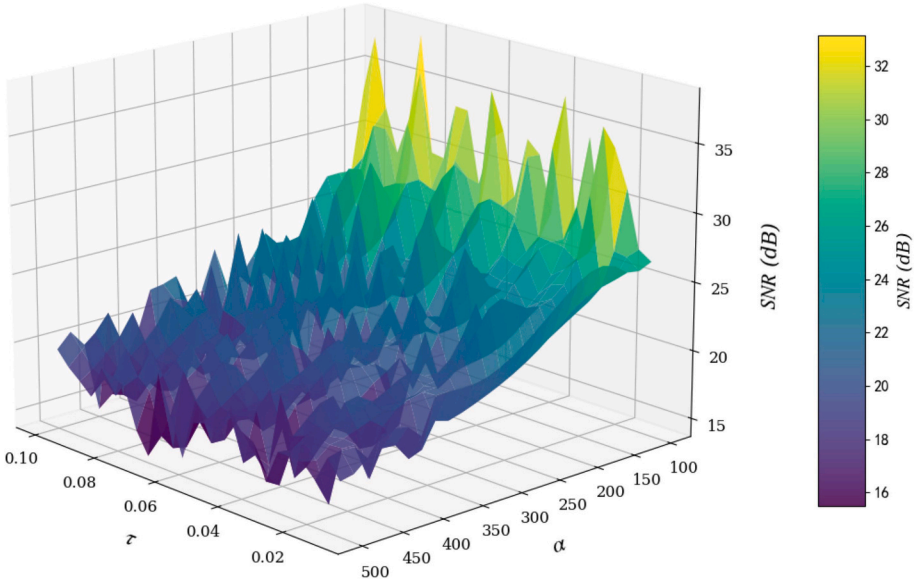


Fig. 5. 3D visualization of PSO-based parameter optimization for VMD ($K = 3$) on the SNR landscape.

Table 3
Performance comparison of the proposed SFT model and single-modal baseline models across different forecast horizons.

Methods	Horizon	SFT	GRU	LSTM	Transformer	ResNet18	MobileNetV2	ViT
MAE (kW)	10 min	0.580	0.690	0.730	0.666	0.841	0.783	0.694
	20 min	0.716	0.935	0.942	1.020	0.996	0.803	0.801
	30 min	0.769	1.091	0.883	0.860	1.020	0.948	0.841
R^2	10 min	0.984	0.977	0.971	0.980	0.981	0.979	0.981
	20 min	0.975	0.969	0.968	0.964	0.971	0.970	0.969
	30 min	0.966	0.961	0.954	0.966	0.958	0.965	0.964
RMSE (kW)	10 min	1.103	1.197	1.346	1.200	1.193	1.149	1.176
	20 min	1.253	1.386	1.402	1.494	1.334	1.367	1.381
	30 min	1.457	1.554	1.675	1.560	1.602	1.475	1.500

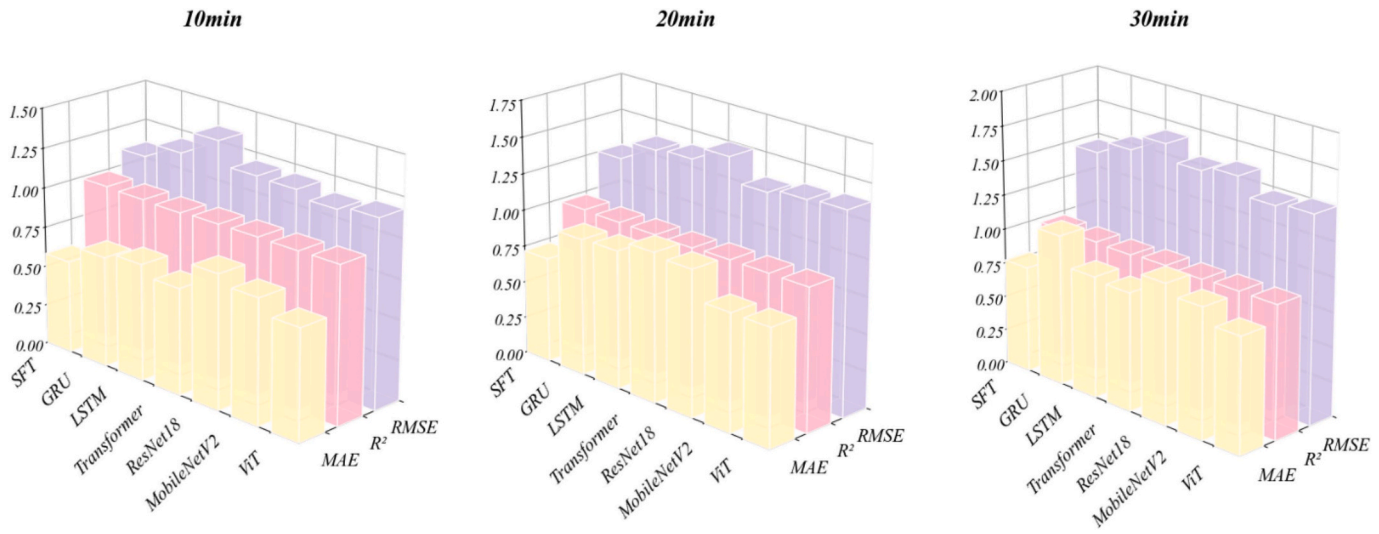


Fig. 6. 3D bar chart of performance differences between the proposed SFT model and single-modal baseline models across different forecast horizons.

Table 4
Performance comparison of the SFT model under different static covariate input scenarios across multiple forecast horizons.

Metrics	Horizon	No Static	Capacity	Location	Both
MAE(kW)	10 min	0.668	0.580	0.685···	0.754
	20 min	0.847	0.716	0.867	0.825
	30 min	0.783	0.769	0.887	1.109
RMSE(kW)	10 min	1.143	1.103	1.178	1.342
	20 min	1.539	1.253	1.320	1.330
	30 min	1.429	1.457	1.570	1.834
R^2	10 min	0.974	0.984	0.972	0.964
	20 min	0.953	0.975	0.965	0.965
	30 min	0.959	0.966	0.951	0.933

increases beyond this point resulte in performance degradation. Therefore, $K = 3$ is selected. The corresponding penalty factor $\alpha = 213.83$ and step size $\tau = 0.016$ are obtained. This set of optimal parameters is applied to decompose and reconstruct the original power sequence, serving as input features for subsequent forecasting models. The three-dimensional distribution of parameter search under these conditions is shown in Fig. 5.

4.2. Benchmark evaluation

To evaluate the overall performance of the SFT model, this study compares it with both temporal and visual models. Table 3 presents the performance comparison between the SFT model and various single-modal baseline models across different forecasting horizons. Fig. 6 visually illustrates the performance differences corresponding to each forecasting horizon using a three-dimensional bar chart. In the 10-minute forecasting task, SFT achieves an MAE of 0.580 kW and an R^2 of 0.984, demonstrating the best overall performance. Compared to the optimal temporal model Transformer, SFT reduces MAE by 12.9%, validating the importance of incorporating image information for capturing short-term power fluctuations.

When the forecasting horizon extends to 20 min, SFT's advantage further expands, reducing MAE by 10.6% compared to ViT and achieving over 23% reductions compared to both GRU and LSTM, highlighting the limitations of unimodal models in handling complex mid-term variations. In 30-minute forecasts, SFT maintains the lowest MAE, reducing it by 10.6% compared to Transformer and 32.6% compared to ResNet18. The results demonstrate that Swin Transformer's feature extraction capabilities play a pivotal role across different forecast horizons, underpinning the exceptional robustness of the SFT model.

4.3. Ablation studies

4.3.1. Static feature ablation

For the TFT model, the selection of static features also significantly impacts forecasting performance. As shown in Table 4, the model achieves optimal performance when only power plant capacity is used as a static covariate. This trend is further shown in Fig. 7, which compares the forecasting performance under different static-feature configurations. When geographic location information is introduced as a static covariate, the forecasting performance degrades. Specifically, both MAE and RMSE increase, while the R^2 decreases across the forecasting horizons, indicating that location features may introduce redundancy or noise rather than providing additional predictive value. Therefore, in subsequent experiments, the SFT model retains only power plant capacity as the input feature.

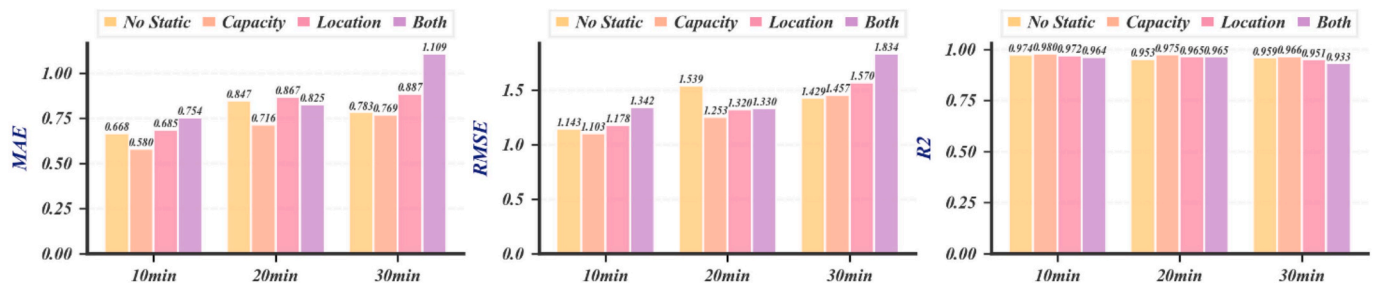


Fig. 7. Bar chart comparison of SFT model forecasting performance under different static covariate input scenarios.

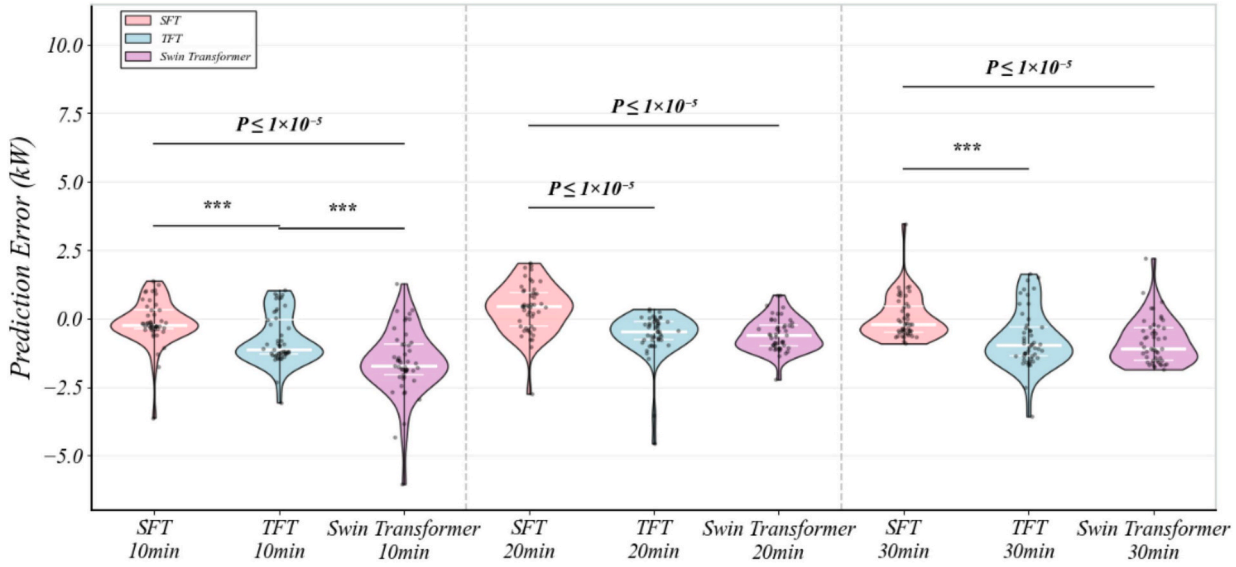


Fig. 8. Violin plot comparison of forecast error distributions for the SFT model and its single-modal components.

Table 5
Performance comparison of the proposed SFT model and its single-modal component models in the ablation study.

Metric	Horizon	SFT	Swin Transformer	TFT
MAE (kW)	10 min	0.580	0.886	0.685
	20 min	0.716	0.818	0.975
	30 min	0.769	1.095	0.830
RMSE (kW)	10 min	1.103	1.247	1.300
	20 min	1.253	1.178	1.320
	30 min	1.457	1.556	1.500
R^2	10 min	0.984	0.975	0.973
	20 min	0.975	0.977	0.972
	30 min	0.966	0.960	0.963

4.3.2. Single-modal component ablation

To dissect the contributions of multimodal fusion mechanisms, this study conducts ablation experiments comparing SFT with pure temporal TFT and pure visual Swin Transformer models. As observed in Fig. 8, SFT exhibits the smallest deviation and highest concentration in error distributions across all forecasting tasks, demonstrating its outstanding robustness and stability. The corresponding quantitative results and statistical comparisons are summarized in Table 5. In contrast, TFT and Swin Transformer exhibit broader error distributions with pronounced tail extensions, indicating their forecasts are more susceptible to perturbations in complex scenes. As the forecast horizon extends from short-term to longer intervals, the error distributions of all three methods generally become more dispersed, with overall forecast stability declining. However, SFT maintains a relatively compact distribution pattern, demonstrating its continued robustness advantage in long-term tasks. Significance tests reveal that error differences among the three methods reach highly significant levels in short-term tasks. This conclusion aligns closely with the visual characteristics of the error distributions in the figure, further validating the effectiveness of multimodal fusion strategies in enhancing both accuracy and stability.

4.3.3. Backbone replacement ablation

This section evaluates the impact of combining time-series and image components in the SFT model through backbone network replacement experiments. Results are documented in Table 6, while Fig. 9 illustrates the power curve comparison for a typical forecast day on September 27. The shaded regions in the figure represent 95% confidence intervals (CI), which reflect the prediction uncertainty and reliability of each model. Narrower confidence intervals indicate more stable and consistent forecasting performance. The SFT model exhibits noticeably narrower confidence intervals across most time steps.

In terms of time-series modeling, the SFT model achieves the lowest MAE in the 10-minute and 30-minute forecasts, at 0.580 kW and 0.769 kW, respectively, outperforming GRU-Swin Transformer and Transformer-Swin Transformer. Although GRU-Swin Transformer obtains the best MAE in the 20-minute forecast, SFT reduces the MAE by approximately 15.7% compared with GRU-Swin Transformer in the 30-minute forecast, demonstrating its robustness in longer-horizon forecasting.

In the image branch, SFT also delivers the best MAE in the 10-minute and 30-minute forecasts among the TFT-based image-backbone variants. In the 10-minute forecast, SFT reduces MAE by approximately 7.6% and 0.7% compared with TFT-ViT and TFT-MobileNetV2, respectively. While TFT-MobileNetV2 achieves the lowest MAE in the 20-minute

Table 6
Performance comparison of different time-series–image model combinations in the backbone replacement experiment.

Method	Horizon	SFT	GRU-Swin Transformer	Transformer-Swin Transformer	TFT-MobileNetV2	TFT-ViT
MAE (kW)	10 min	0.580	0.780	0.891	0.584	0.628
	20 min	0.716	0.632	0.699	0.643	1.051
	30 min	0.769	0.912	0.906	0.907	0.935
RMSE (kW)	10 min	1.103	1.280	1.292	1.182	1.151
	20 min	1.253	1.184	1.236	1.400	1.819
	30 min	1.457	1.520	1.710	1.849	1.854
R^2	10 min	0.984	0.973	0.973	0.977	0.978
	20 min	0.975	0.977	0.975	0.968	0.946
	30 min	0.966	0.962	0.953	0.945	0.944

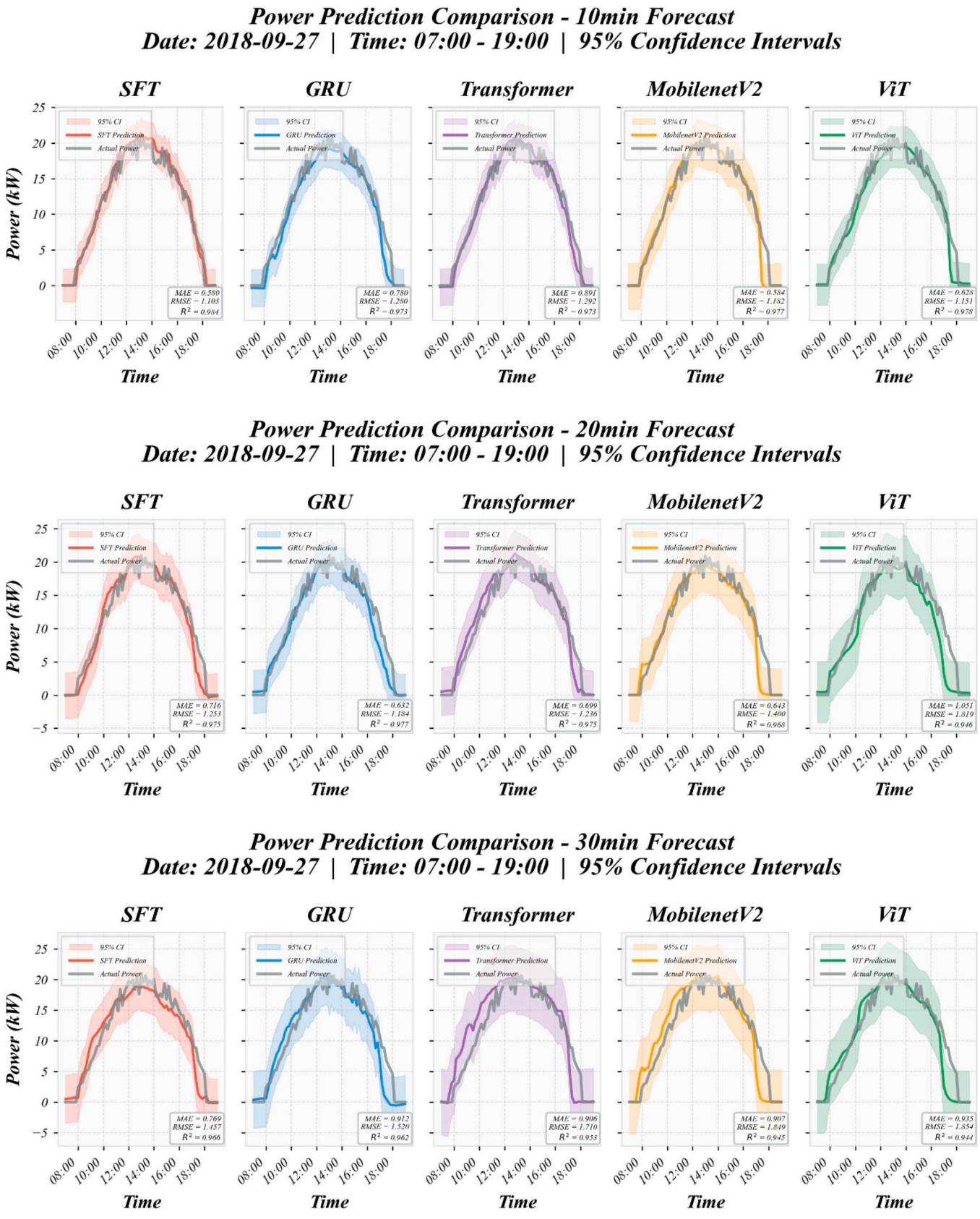


Fig. 9. Comparison of forecasted and actual power curves for different backbone combinations on a typical forecasting day. Note: "GRU/Transformer" replace the temporal branch (TFT) with the image branch fixed as Swin Transformer, whereas "MobileNetV2/ViT" replace the image branch (Swin Transformer) with the temporal branch fixed as TFT.

Table 7
Key hyperparameter configurations of different cross-modal fusion strategies.

Fusion Strategy	Key Configurations
Early Fusion	LSTM hidden size 128, 8 attention heads, key dimension 16.
Mid Fusion	FAF with 4 attention heads, key dimension 32, and projection dimension 128.
Late Fusion	Gated Fusion with gate dimension 1 and softmax normalization.

Table 8
Performance comparison of different cross-modal fusion strategies across multiple forecast horizons.

Metric	Horizon	Late fusion	Early fusion	Mid-fusion
MAE (kW)	10 min	0.580	0.705	0.719
	20 min	0.716	0.805	0.928
	30 min	0.769	1.025	1.060
RMSE (kW)	10 min	1.103	1.308	1.189
	20 min	1.253	1.432	1.631
	30 min	1.457	1.847	1.574
R^2	10 min	0.984	0.972	0.977
	20 min	0.975	0.967	0.957
	30 min	0.966	0.945	0.960

forecast among the image-backbone variants, SFT regains the lead in the 30-minute forecast, reducing MAE by approximately 17.8% and 15.2% compared with TFT-ViT and TFT-MobileNetV2, respectively.

4.3.4. Fusion strategy ablation

To investigate the performance differences among various cross-modal fusion strategies, this section compares three approaches: early fusion, mid-fusion, and late fusion. Quantitative performance metrics are presented in Table 8, while core hyperparameter configurations are detailed in Table 7. Early fusion directly concatenates raw features $F_{early} = [X_v; X_t]$ at the front end into a unified encoder; mid-fusion employs cross-attention $F_{mid} = FAF(X_v, X_t)$ for deep interaction after independent encoding feature extraction.

Overall results demonstrate that late fusion exhibits significant advantages across multiple evaluation metrics. In the scatter plot of ground truth versus forecasted values shown in Fig. 10, forecasts from late fusion cluster closely near the diagonal line, indicating high consistency with reference values. In contrast, early and mid-fusion methods show noticeably dispersed points in the high-power region, deviating more significantly from the diagonal line, reflecting their insufficient forecasting accuracy. Quantitative results in Table 8 further confirm this observation. Averaged over the three forecast horizons, late fusion reduces MAE by 18.5% and 23.7% compared with early fusion and mid-

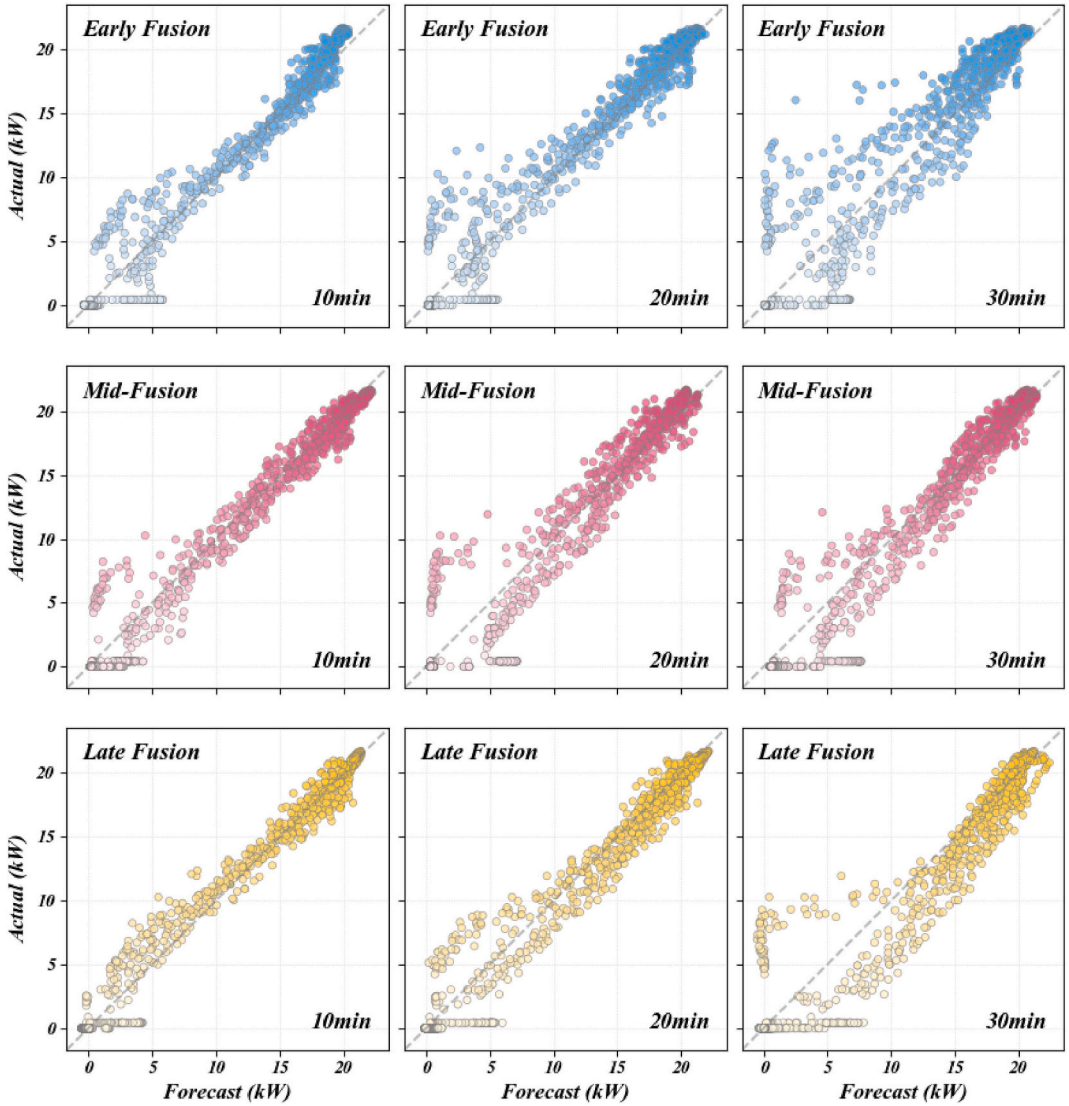


Fig. 10. Scatter plot comparison of forecasted vs. actual values for different cross-modal fusion strategies.

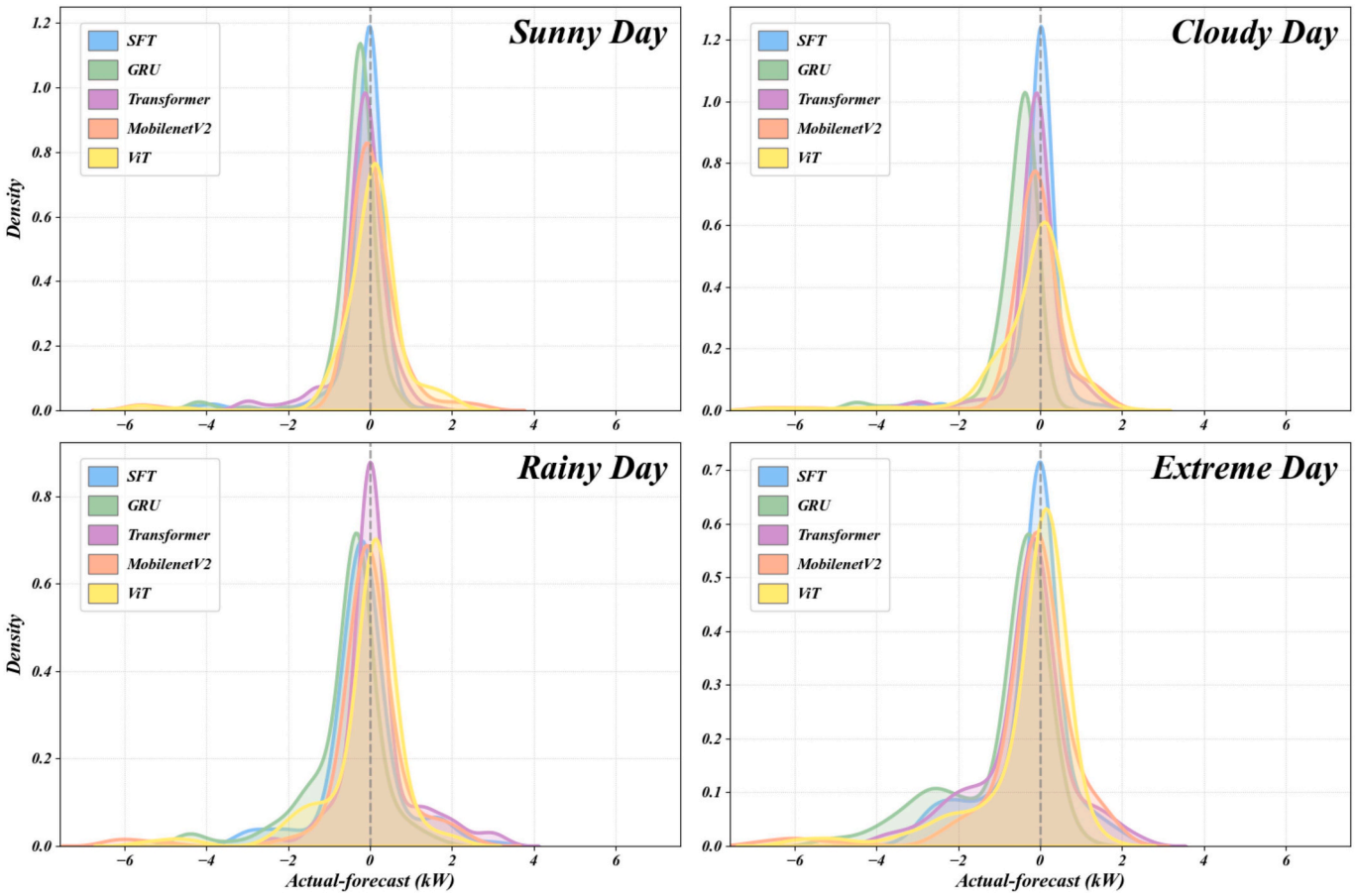


Fig. 11. Kernel density estimation of forecast error distributions for different models under complex weather condition. Note: "GRU/Transformer" replace the temporal branch (TFT) with the visual branch fixed as Swin Transformer, whereas "MobileNetV2/ViT" replaces the visual branch (Swin Transformer) with the temporal branch fixed as TFT.

fusion, respectively. The corresponding average RMSE reductions are 16.9% and 13.3%, while the average R^2 increases by approximately 1.4 and 1.0 percentage points, respectively. These results indicate that late fusion provides more accurate and stable forecasts across different prediction horizons.

Mechanistic analysis further indicates that early fusion suffers from encoding interference due to direct concatenation of heterogeneous features, impairing overall modeling efficiency; mid-fusion maintains good accuracy for short-term tasks but exhibits insufficient stability in cross-modal weights during more complex forecasting scenarios; late fusion fully extracts features from each modality through independent encoding, then applies adaptive weighting via a gated-attention mechanism. This effectively preserves modality uniqueness while optimizing fusion decisions. This design ensures late fusion delivers higher stability and environmental adaptability across diverse forecasting tasks.

4.4. Generalization performance

Regarding the backbone network replacement experiment results, this study combines kernel density estimation to analyze the generalization performance of different multimodal structures under various weather conditions, with results shown in Fig. 11. The overall trend indicates that SFT exhibits the highest error concentration in clear environments, with a peak density of approximately 1.2 and a narrow bandwidth, leading in both stability and forecasting accuracy. In contrast, when encountering more complex scenarios such as cloud cover changes, rainfall, or extreme weather, the overall performance of all architectures declines. Error distributions become more dispersed and exhibit long tails, reflecting increased forecast uncertainty. Notably, alternative architectures like GRU-Swin Transformer, Transformer-Swin Transformer, TFT-MobileNetV2, and TFT-ViT generally exhibit lower peak densities than SFT and broader distribution ranges. Comprehensive analysis reveals that even under significantly deteriorated environmental conditions, SFT maintains high concentration and short-tailed distribution characteristics, indicating its superior stability and environmental adaptability compared to other architectures.

5. Conclusion

As global demand for renewable energy grows, the accuracy of photovoltaic power forecasting is critical for energy dispatch and grid stability. Addressing the limitations of traditional unimodal approaches in capturing complex weather patterns, this study proposes a multimodal model integrating sky imagery with historical meteorological data. Experimental results demonstrate that the model achieves a MAE of 0.580 kW and an R^2 of 0.984 for 10-minute forecasts, with R^2 values of 0.975 and 0.966 for 20-minute and 30-minute forecasts, respectively. Its performance surpasses that of comparison models, showcasing exceptional accuracy. Through cross-modal feature interaction, the model fully leverages cloud texture and meteorological temporal characteristics, significantly enhancing its ability to capture complex weather patterns. Its innovative fusion mechanism strengthens modal synergy, providing a reliable tool for photovoltaic system optimization.

Despite these significant findings, future improvements are possible. On one hand, incorporating datasets from larger-capacity PV power plants is recommended to validate the model's generalization capabilities under diverse and extreme weather conditions. On the other hand, optimizing feature fusion strategies—such as exploring adaptive weight

allocation mechanisms—could further enhance accuracy. Additionally, integrating high-resolution external data or more complex weather pattern analysis would broaden the model's applicability. These improvements will advance the practical deployment of multimodal models, delivering more robust solutions for high-precision PV power forecasting.

CRediT authorship contribution statement

He Jiang: Writing – original draft, Methodology, Conceptualization. **Zehua Yang:** Writing – review & editing, Software, Formal analysis. **Yushuai Li:** Writing – review & editing, Data curation. **Rui Wang:** Validation. **Jiaqi Shi:** Supervision. **Yan Zhao:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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